Early Detection of Plant Nutritional Deficiency via Spectral Analysis of Roots and Leaves

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***Abstract*—The early detection of nutrient deficiencies in plants is a key factor in sustainable and resource-efficient agriculture. However, it remains unclear to what extent changes in the root area can also be detected early using multispectral imaging (MSI). The objective of this study is to systematically investigate whether root-based spectral diagnostics allow for earlier identification of plant health issues compared to traditional leaf-based assessment under controlled conditions. A custom-built mounting frame was developed, allowing for daily imaging of the plants and their root systems over a period of seven consecutive days. This multimodal imaging approach was selected to provide complementary visual information suitable for subsequent scientific analysis. The findings could not only contribute to the further development of existing sensor technologies, but also open up new avenues for sensor-based decision support systems in plant production.**

***Index Terms*—Multispectral Imaging, Plant Stress Detection, Nutrient Deficiency, Root Spectroscopy, Precision Agriculture, Spectral Analysis, Non-invasive Diagnostics, Hydroponics, Lettuce (*Lactuca sativa*), Early Plant Health Monitoring**

I. Introduction

The early detection of nutrient deficiencies in plants is a key factor in sustainable and resource-efficient agriculture. A deficiency in essential nutrients can significantly impair plant growth, metabolism, and yield. Against the backdrop of increasing demands for efficiency and environmental compatibility, the use of precise, preferably non-invasive diagnostic methods in plant production is becoming increasingly important. The aim is to detect physiological changes at an early stage so that targeted measures can be taken in good time.

Multispectral cameras have established themselves as a promising technology in plant phenotyping in recent years. By capturing reflection patterns in different wavelength ranges of light, they provide insight into biochemical and structural changes in plant issue, often before visible symptoms appear. Most of the research to date has focused on leaves as a diagnostic target. However, it remains unclear to what extent changes in the root area can also be detected early using multispectral imaging. Possibly even earlier than in the leaves.

Research Gap and Objective

Despite its growing use in agricultural research, MSI has rarely been applied to root tissues for nutrient deficiency detection. Several critical questions remain unanswered: Can reflectance signals from root zones be reliably captured using MSI? Which spectral bands are most sensitive to nutrient-induced root changes? And can root-based signals serve as earlier indicators than those detected in leaves?

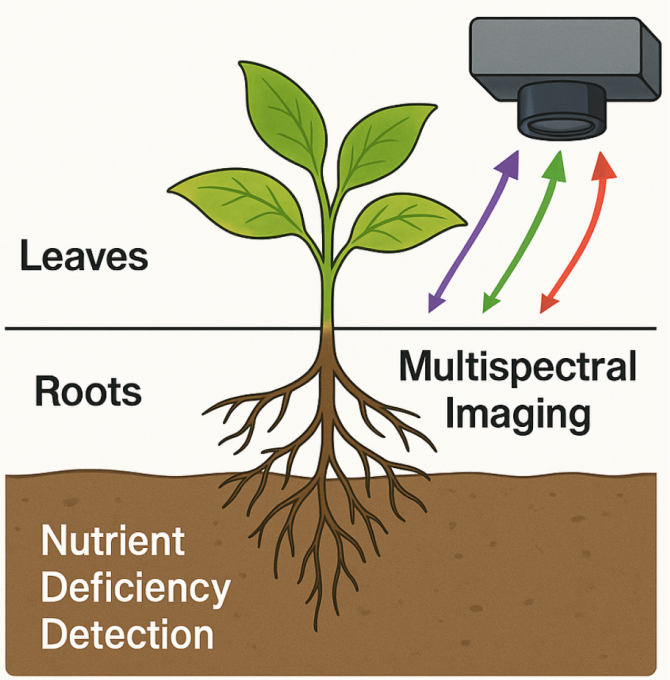

Fig. 1. Concept of Detection from Leaf vs. Roots

Addressing these questions could significantly improve the diagnostic scope of multispectral sensing technologies and open new avenues for sensor-based decision support systems in plant production. Therefore, the objective of this study is to evaluate the potential of MSI for early detection of nutrient deficiencies in the root zone, and to compare the temporal and spectral sensitivity of root-based versus leaf-based diagnostics under controlled conditions.

A. Motivation

The root is the central interface between the plant and the soil and is therefore the primary site for nutrient uptake. It therefore seems obvious that the first signs of a deficiency will initially appear in root development or physiology before above-ground symptoms become apparent. The assumption

that multi-spectral imaging is capable of revealing even subtle changes in the root area raises interesting questions, both with regard to fundamental plant physiological processes and in the context of practical application in precision agriculture.

Such early and location-specific diagnostics would have the potential to enable more targeted fertilization recommendations, use resources more efficiently, and reduce environmental impact. Against this background, the present study aims to systematically investigate whether nutrient deficiencies in plants can be detected earlier in the root area than in the leaf area using multi-spectral imaging. The findings could not only contribute to the further development of existing sensor technologies, but also open up new approaches for sensor-based monitoring of crops.

II. State of the Art

Hyperspectral imaging has emerged as a powerful and non-invasive method of analyzing the root system in hydroponically cultivated crops such as lettuce (*Lactuca sativa*), spinach, and basil under controlled conditions [1]–[3]. Modern pushbroom hyperspectral cameras can capture hundreds of spectral bands in the visible to near-infrared range (500–1700 nm) [2], [3], making them suitable for characterizing fine structural and physiological differences in root tissues. In transparent nutrient film technique (NFT) systems, root zones are periodically scanned throughout the day, where each pixel contains a complete spectral signature [1]. Data calibration is typically performed using white and dark reference frames, followed by spectral preprocessing steps such as log-linear transformation and asymmetric least squares correction [3].

Bodner et al. [3] proposed an image processing pipeline in which root segmentation is achieved using multithresholding or fuzzy clustering on calibrated hyperspectral data. Their findings indicated that hyperspectral root segmentation on bright backgrounds was as accurate as RGB-D based methods. Alternatively, specific spectral features and indices are extracted from segmented root spectra. Jin et al. [1] were able to classify root aging in spinach based on reflectance ratios at 498 nm and 601 nm, while Antony et al. [2] introduced a "Root Health Index" using SWIR signatures and spectral angle mapping to assess root vitality in hydroponically grown lettuce.

One major challenge remains the spectral overlap between water, nutrient solution, and organic matter, particularly due to strong water absorption bands. Transparent growth containers, however, reduce occlusion by growth media and improve root visibility [1]. Overall, these studies demonstrate that hyperspectral imaging combined with robust image analysis techniques enables efficient and non-destructive monitoring of root development and stress dynamics in hydroponic systems [1]–[3].

III. Concept Description

A. Objective

The aim of this project is to investigate whether the root system of plants exhibits spectral changes due to nutrient deficiency earlier than the leaves, enabling earlier diagnosis of plant health degradation. This hypothesis is tested through a controlled experiment using a spectralcamera to analyze the spectral characteristics of both roots and leaves in Lactuca sativa (garden lettuce).

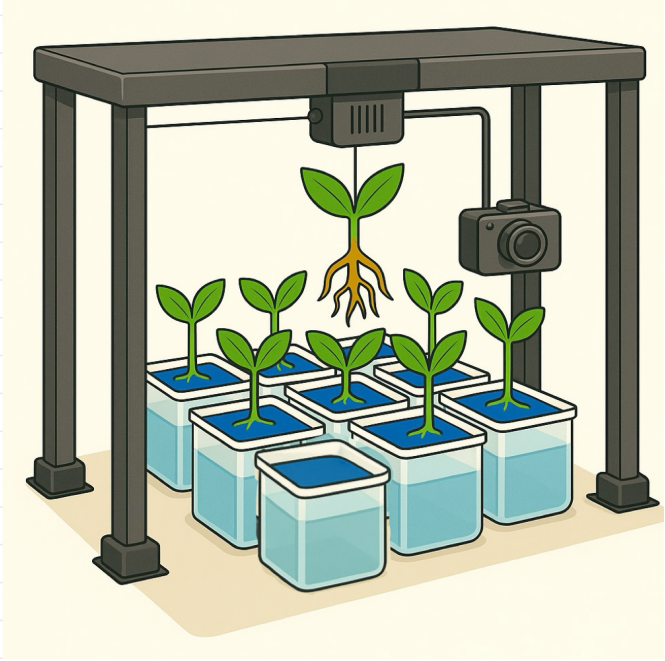

Fig. 2. Mechanical Construction

B. Scientific Background

Nutrient deficiency is a major stress factor in crop production. Typically, its diagnosis relies on visible symptoms in the leaves, such as discoloration or deformation. However, these symptoms occur at a relatively late stage of physiological stress. Since roots are the primary interface for nutrient uptake, they may show biochemical and structural responses earlier. Spectral analysis, particularly in the visible to near-infrared (400-1000 nm) range, has proven to be a powerful non-invasive method to detect internal plant stress responses. This project explores the hypothesis that root-based spectral diagnostics allow for earlier identification of plant health issues compared to traditional leaf-based assessment.

C. Experimental Design

As shown in Figure 3, nine genetically identical Lactuca sativa plants are divided into two groups:

- Reference group (3 plants): cultivated in nutrient-rich hydroponic solution
- Test group (6 plants): cultivated in distilled water to induce nutrient deficiency

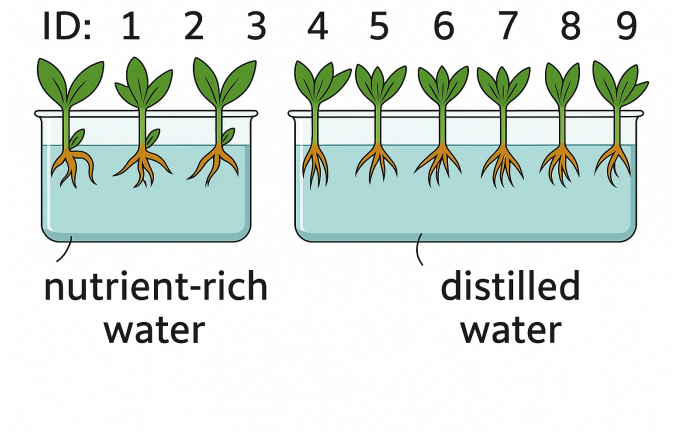

Fig. 3. Comparison of plant growth in nutrient-rich vs. distilled water

Over a period of ten days, daily spectral measurements are taken from both roots and leaves using a calibrated VIS-NIR spectrometer. All measurements are standardized with respect to light, distance, and ambient conditions. Observations are recorded in a structured Excel spreadsheet for documentation and reproducibility.

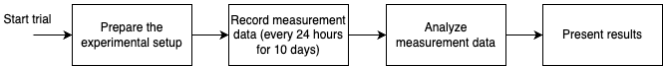

Fig. 4. Experimental Procedure

Fig. 5. Real plants in pots

D. Experimental Setup

The experimental setup was realized in a 3×3 grid comprising three healthy plants alongside six plants subjected to nutrient deficiency or stress conditions. This arrangement was specifically chosen to enable systematic comparability between the healthy control group and the stressed or nutrient-limited specimens.

To document the physiological responses over time, a custom-built mounting frame was developed, allowing for daily imaging of the plants and their root systems over a period of seven consecutive days. During image acquisition, the plants were temporarily lifted out of the nutrient solution to minimize light refraction and reflections caused by the water surface.

For enhanced data quality and reference, two types of images were captured: monochrome and RGB photographs. The monochrome images were acquired using a **LUCID Vision Labs Phoenix™ PHX050S-MC** camera, whereas the RGB photographs were taken with an **iPhone 13 mini**. This multimodal imaging approach was selected to provide complementary visual information suitable for subsequent scientific analysis.

IV. Protocol for Acquiring Reference Images of Test Plants

The following protocol describes the standardized procedure for capturing high-quality reference images of test plants under controlled conditions. This ensures reproducibility and consistency across measurements.

A. Equipment and Measurement Details

- Device: LUCID Vision Labs Phoenix™ PHX050S-MC (400–1000 nm)
- Device 2: iPhone 13 mini (400-700 nm)
- Tissues measured: One fixed leaf per plant, cleaned roots
- Measurement frequency: Once per day, same time daily
- Data format: CSV files with metadata logged in an Excel template
- Calibration: White reference scan before every session

Fig. 6. Test setup for taking pictures

B. Brief Protocol

1. Open the imaging tool and ensure the camera is active. Turn off ambient light, close blinds, and set LED lighting to 6500 K.
2. Position the mobile phone camera approximately 2 cm above the sample using 2× zoom, aligned squarely and slightly tilted.
3. Capture reference images with and without the plant; check image quality.
4. Record the timestamp in the Excel file on a separate device.
5. Remove the plant, wait about 20 seconds, then capture images (first with Mac, then phone). Repeat this process for all plants.
6. Measure and log water nutrient concentration, ambient temperature, and humidity in Excel. Check distilled water quality and document the results.
7. Update the master file with environmental data; adjust water nutrients if needed to remain within optimal range.
8. Transfer approximately 100 images from the phone, organize them into folders, and compress into ZIP archives.
9. Turn off the camera and secure the equipment.

C. Documentation and Reproducibility

To ensure reproducibility, all measurements and environmental conditions were carefully standardized and consistently documented. This includes the plant ID, date and time, time since root exposure, pH value (via indicator strips), ambient temperature, humidity, and observations. The plants originate from a single cultivation batch, are in the same developmental stage, and are grown hydroponically in identical mesh pots without residual soil. Hyperspectral image acquisition takes place at a fixed time of day (19:00) in a fully darkened room to eliminate fluctuations in ambient light. Illumination is provided exclusively by a fixed LED light source with a defined color temperature. The distance, angle and orientation of the camera are maintained through a custom-built mounting fixture.

Each plant is recorded from ten perspectives, five images of the leaves and five of the roots, by rotating the pot in 60° increments. The rotation starts from a fixed reference point and is guided by color-coded markings on the pots. Positioning is achieved through visual alignment of the markings with a fixed reference on the holder, enabling consistent multi-angle acquisition. A white and dark reference calibration is performed prior to each measurement session. Furthermore, the time elapsed (in seconds) since the roots were lifted from the nutrient solution is documented for each image to account for the potential dehydration effects during imaging.

In addition, environmental parameters such as the pH value (measured using indicator strips), room temperature, and relative humidity are recorded for each session. (Optionally, electrical conductivity [EC] can be added to quantify nutrient concentrations in future experiments.) The plants are imaged in the same order during each session. This standardized protocol ensures high temporal, technical, and spatial reproducibility and provides a solid foundation for analyzing spectral changes in response to nutrient stress over time.

D. Objective

V. Preliminary Results and Observations

After seven days of observation, no significant spectral differences between the nutrient-rich control group and the nutrient-deficient test group could be identified in the monochrome images. Minor visual indicators of plant stress, such as downward leaf curling, are visible in both monochrome and RGB images. However, the added value

of the monochrome imaging system is questionable, as these changes are equally or more clearly visible in standard RGB photographs.

The camera used in this study operates without spectral resolution across defined wavelength bands. As it captures only grayscale intensities under broad-spectrum white illumination, it does not allow for wavelength-specific analysis of reflectance properties. As a result, spectral effects such as pigment degradation or water-related absorption features cannot be isolated or tracked over time. This limitation prevents the detection of potential early stress markers in the root zone that might have manifested in specific parts of the spectrum.

Environmental measurements confirm clear differences in electrical conductivity (EC) between the two solution types, validating the nutrient manipulation. However, the pH level consistently remained above the optimal range for lettuce (5.8–6.2) and had to be manually adjusted using diluted vinegar. The room temperature exceeded the recommended range for lettuce growth during most sessions, and no active oxygenation was provided in the water containers. These environmental factors may contribute to physiological stress and affect the plant's response patterns.

Despite these limitations, the experiment has proceeded without any plant losses. All plants remain alive, and data collection has been consistent. The reproducibility of imaging and measurement conditions ensures that future improvements—such as extended observation periods, refined sensor technology, and improved environmental control—can be meaningfully evaluated against the existing data baseline.

VI. Conclusion

This study set out to investigate whether spectral signals of nutrient deficiency can be detected earlier in the root zone than in the leaf tissue of *Lactuca sativa*. A controlled hydroponic experiment was established, with a standardized imaging and measurement protocol applied over a period of seven days. While no definitive spectral differences were observed in the monochrome imaging data, RGB photographs did indicate slight visual stress symptoms such as leaf drooping. These findings highlight both the importance of wavelength-specific sensing capabilities and the need for an extended observation period. Environmental challenges such as elevated temperature and unstable pH values also emerged as critical factors affecting plant response.

Despite these limitations, the experimental setup proved to be highly reproducible and provides a solid foundation for further investigation. Future work will focus on integrating true multispectral or hyperspectral cameras, improving environmental control, and extending the measurement duration. The presented protocol and findings offer valuable insights for developing early-stage plant diagnostics and sensor-based monitoring systems in precision agriculture.

VII. Acknowledgment

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